

Subnetting

```
R1
R1#wr
Building configuration...
[OK]
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

50.0.0.0/8 is variably subnetted, 6 subnets, 5 masks
S    50.0.0.0/24 [1/0] via 50.0.1.194
S    50.0.1.0/25 [1/0] via 50.0.1.194
C    50.0.1.128/26 is directly connected, Ethernet0/0
L    50.0.1.129/32 is directly connected, Ethernet0/0
C    50.0.1.192/30 is directly connected, Ethernet0/1
L    50.0.1.193/32 is directly connected, Ethernet0/1
R1#
```

```
R2
[OK]
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

50.0.0.0/8 is variably subnetted, 8 subnets, 5 masks
S    50.0.0.0/24 [1/0] via 50.0.1.198
C    50.0.1.0/25 is directly connected, Ethernet0/0
L    50.0.1.1/32 is directly connected, Ethernet0/0
S    50.0.1.128/26 [1/0] via 50.0.1.193
C    50.0.1.192/30 is directly connected, Ethernet0/1
L    50.0.1.194/32 is directly connected, Ethernet0/1
C    50.0.1.196/30 is directly connected, Ethernet0/2
L    50.0.1.197/32 is directly connected, Ethernet0/2
R2#
```

```
R3
[OK]
R3#
*Sep 10 07:16:20.022: %SYS-5-CONFIG_I: Configured from console by console
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

50.0.0.0/8 is variably subnetted, 6 subnets, 5 masks
C    50.0.0.0/24 is directly connected, Ethernet0/0
L    50.0.0.1/32 is directly connected, Ethernet0/0
S    50.0.1.0/25 [1/0] via 50.0.1.197
S    50.0.1.128/26 [1/0] via 50.0.1.197
C    50.0.1.196/30 is directly connected, Ethernet0/1
L    50.0.1.198/32 is directly connected, Ethernet0/1
R3#
```

```

R1
R1(config)#ip dhcp excluded-address 50.0.1.129 50.0.1.140
R1(config)#do wr
Building configuration...
[OK]
R1(config)#ip dhcp pool hr
R1(dhcp-config)#network 50.0.1.128 255.255.255.192
R1(dhcp-config)#default-router 50.0.1.129
R1(dhcp-config)#dns-server 8.8.
*Sep 10 07:57:23.451: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R1(dhcp-config)#dns-server 8.8.8.8
R1(dhcp-config)#exit
R1(config)#do wr
Building configuration...
[OK]

```

```

R2
C      50.0.1.196/30 is directly connected, Ethernet0/2
L      50.0.1.197/32 is directly connected, Ethernet0/2
R2#
R2#
R2#
R2#config t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ip dhcp excluded-address 50.0.1.1 50.0.1.10
R2(config)#do wr
Building configuration...
[OK]
R2(config)#ip dhcp pool production
R2(dhcp-config)#network 50.0.1.0 255.255.255.128
R2(dhcp-config)#default-router 50.0.1.1
R2(dhcp-config)#dns-server 8.8.8.8
R2(dhcp-config)#do wr
Building configuration...
[OK]
R2(dhcp-config)#exit
R2(config)#exit
R2#
*Sep 10 07:14:34.900: %SYS-5-CONFIG_I: Configured from console by console
R2#wr
Building configuration...

```

```

R3
Ethernet0/2          unassigned      YES TFTP    down        down
Ethernet0/3          unassigned      YES TFTP    down        down

R3#
*Sep 10 07:04:33.208: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R3#config t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ip dhcp excluded-address 50.0.0.1 50.0.0.10
R3(config)#do wr
Building configuration...
[OK]
R3(config)#ip dhcp pool sales
R3(dhcp-config)#network 50.0.0.0 255.255.255.0
R3(dhcp-config)#default-router 50.0.0.1
R3(dhcp-config)#dns-server 8.8.8.8
R3(dhcp-config)#do wr
Building configuration...
[OK]
R3(dhcp-config)#exit
R3(config)#exit
R3#wr
Building configuration...

```

```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address . . . . . : 50.0.1.142
    Subnet Mask . . . . . : 255.255.255.192
    Default Gateway . . . . . : 50.0.1.129

C:\Documents and Settings\user>
```

```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address . . . . . : 50.0.0.11
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 50.0.0.1

C:\Documents and Settings\user>
```

```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address . . . . . : 50.0.1.11
    Subnet Mask . . . . . : 255.255.255.128
    Default Gateway . . . . . : 50.0.1.1

C:\Documents and Settings\user>
```

```
VPC-1
Modified version for EVE-NG.

Press '?' to get help.

Executing the startup file

VPC-1> ip dhcp
DDORA IP 50.0.1.141/26 GW 50.0.1.129

VPC-1> show ip

NAME       : VPC-1[1]
IP/MASK    : 50.0.1.141/26
GATEWAY    : 50.0.1.129
DNS        : 8.8.8.8
DHCP SERVER : 50.0.1.129
DHCP LEASE  : 86385, 86400/43200/75600
MAC        : 00:50:79:66:68:ac
LPORT      : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500

VPC-1>
```

```
VPC-2
VPC-2> show ip dhcp

show ip [all]
  Show IPv4 details for including:
  VPC Name, IP address, mask, gateway, DNS, MAC, lport, rhost:rport and MTU.
  (reduced view in tabular format if 'all' option used)

VPC-2> ip dhcp
DDORA IP 50.0.1.12/25 GW 50.0.1.1

VPC-2> show ip

NAME       : VPC-2[1]
IP/MASK    : 50.0.1.12/25
GATEWAY    : 50.0.1.1
DNS        : 8.8.8.8
DHCP SERVER : 50.0.1.1
DHCP LEASE  : 86390, 86400/43200/75600
MAC        : 00:50:79:66:68:ad
LPORT      : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500

VPC-2>
```

```
VPC-3
Modified version for EVE-NG.

Press '?' to get help.

Executing the startup file

VPC-3> ip dhcp
DDORA IP 50.0.0.12/24 GW 50.0.0.1

VPC-3> show ip

NAME       : VPC-3[1]
IP/MASK    : 50.0.0.12/24
GATEWAY    : 50.0.0.1
DNS        : 8.8.8.8
DHCP SERVER : 50.0.0.1
DHCP LEASE  : 86319, 86400/43200/75600
MAC        : 00:50:79:66:68:ae
LPORT      : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500

VPC-3>
```